

بناء صف يمثل **نقطة** فيه الاحداثيات
ويرث منه الصف **دائرة** فيه نصف القطر وتابع الطباعة وتابع
المساحة والمحيط
ويرث منه الصف **أسطوانة** فيه ارتفاع الأسطوانة وحجم ومساحة
كلية وطباعة

```
#pragma once
class Point
{
protected:
    int x, y;
public:
    Point(int = 0, int = 0);
    void setX(int);
    int getX();
    void setY(int);
    int getY();
    void print();
};

#include "Point.h"
#include <iostream>
using namespace std;
Point::Point(int a, int b){ setX(a); setY(b);}

void Point::setX(int a){    x = a;}

int Point::getX(){    return x;}

void Point::setY(int b){    y = b;}

int Point::getY(){    return y;}

void Point::print()
{
    cout << "[ " << getX() << " , " << getY() << " ]" ;
}

#pragma once
#include "Point.h"
class Circle :    public Point
{
protected:
    double r;
public:
    Circle(int = 0, int = 0, double = 0); //Constructor
    void setR(double);
```

```

        double getR();
        double area();
        double cir();
        void print();
};

```

```

#include "Circle.h"
#include <iostream>
using namespace std;
Circle::Circle(int a, int b, double c)
{
    x = a; y = b; r = c;
}
void Circle::setR(double c){    r = c;}
double Circle::getR(){    return r;}

double Circle::area(){    return 22/7*r*r;}

double Circle::cir(){    return 22 / 7 * r*2;}
void Circle::print()
{
    Point::print(); cout << " Radius is " << getR();
}

```

```

#pragma once
#include "Circle.h"
class Cylinder : public Circle
{
    double h;
public:
    Cylinder(int = 0, int = 0, double = 0, double = 0);
    void setH(double);
    double getH();
    double FullArea();
    double Volume();
    void print();
};

```

```

#include "Cylinder.h"
#include <iostream>
using namespace std;
Cylinder::Cylinder(int a, int b, double c, double d)
{
    x = a; y = b; r = c; h = d;
}
void Cylinder::setH(double d){    h = d;}
double Cylinder::getH(){    return h;}
double Cylinder::FullArea(){    return cir()*h+area()*2;}

```

```

double Cylinder::Volume(){ return area()*h;}

void Cylinder::print(){
    Circle::print();
    cout << "\tHeight is " << getH() << endl;
}

```

```

#include<iostream>
#include "Cylinder.h"
using namespace std;

void main()
{
    Point point(12, 14);
    cout << "X-Coordinate is: " << point.getX() << endl;
    cout << "Y-Coordinate is: " << point.getY() << endl;
    point.print();
    //=====
    Circle circle(11,15,10.5);
    circle.print();
    cout << "Area is: " << circle.area() << endl;
    cout << "Circumference is: " << circle.circumference() << endl;
    //=====
    Cylinder cy(19, 15, 11.9, 14.1);
    cy.print();
    cout << "Full Area is: " << cy.FullArea() << endl;
    cout << "Volume is: " << cy.volume() << endl;
}

```